Problem 1. Which of these sentences are propositions? What are the truth values of those that are propositions?

a) Boston is the capital of Massachusetts.
b) Miami is the capital of Florida.
c) $2 + 3 = 5$.
d) $5 + 7 = 10$.
e) $x + 2 = 11$.
f) Answer this question.

Answer. The statements from a to d are valid propositions. However, not all of them are true.

a) True
b) False, the capital of FL is Tallahassee.
c) True
d) False, $5 + 7 = 12$
e) Not a proposition since we don't know the value of x.
f) Not a proposition, it is an instruction.

Problem 2. Let p, q, and r be the propositions:

- p: You have the flu.
- q: You miss the final examination.
- r: You pass the course.

Express each of these propositions as an English sentence.

a) $p \to q$
b) $\neg q \leftrightarrow r$
c) $q \to \neg r$
d) $p \lor q \lor r$
e) $(p \to \neg r) \lor (q \to \neg r)$
f) $(p \land q) \lor (\neg q \land r)$

Answer.

a) You miss the final examination if you have the flu.
b) You will pass the course if and only if you don't miss the final examination.
c) If you miss the final examination, you will not pass the course.
d) You have the flu, or you miss the final examination, or you pass the course.
e) If you have the flu and you miss the final examination, then you do not pass the course.
   - simplified: $(p \land q) \to \neg r$ (work below)
f) You have the flu and you miss the final examination, or you don't miss the final examination and you pass the course.

Work.

Showing that $(p \to \neg r) \lor (q \to \neg r) \equiv (p \land q) \to \neg r$:

We can substitute $p \to \neg r$ and $q \to \neg r$ and the other way around since $p \to q \equiv \neg p \lor q$.

$$\begin{aligned}
&(p \to \neg r) \lor (q \to \neg r) & \\
&(\neg p \lor \neg r) \lor (\neg q \lor \neg r) & \text{(substitution)} \\
&\neg p \lor \neg q \lor \neg r & \\
&\neg(p \land q) \lor \neg r & \neg p \lor \neg q \equiv \neg(p \land q) \\
&(p \land q) \to \neg r & \text{(substitution)}
\end{aligned}$$

Problem 3. Let p, q, and r be the propositions:

- p: You get an A on the final exam.
- q: You do every exercise in this book.
- r: You get an A in this class

Write these propositions using p, q, and r and logical connectives (including negations).

a) You get an A in this class, but you do not do every exercise in this book.
b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
c) To get an A in this class, it is necessary for you to get an A on the final.
d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

Answer.

a) $r \wedge \neg q$
b) $p \wedge q \wedge r$
c) $r \to p$
d) $p \wedge \neg q \wedge r$
e) $(p \wedge q) \to r$
f) $(p \vee q) \leftrightarrow r$

Problem 4. Determine whether these biconditionals are true or false.

a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.
b) $1 + 1 = 2$ if and only if $2 + 3 = 4$.
c) $1 + 1 = 3$ if and only if monkeys can fly.
d) $0 > 1$ if and only if $2 > 1$.

Answer.

a) True; $2 + 2 = 4 \equiv \top$, $1 + 1 = 2 \equiv \top$, $\top \leftrightarrow \top = \top$
b) False; $1 + 1 = 2 \equiv \top$, $2 + 3 = 4 \equiv \bot$, $\top \leftrightarrow \bot = \bot$
c) True; $1 + 1 = 3 \equiv \bot$, monkeys can fly $\equiv \bot$, $\bot \leftrightarrow \bot = \top$
d) False; $0 > 1 \equiv \bot$, $2 > 1 \equiv \top$, $\bot \leftrightarrow \top = \bot$

Problem 5. Determine whether each of these conditional statements is true or false.

a) If $1 + 1 = 3$, then unicorns exist.
b) If $1 + 1 = 3$, then dogs can fly.
c) If $1 + 1 = 2$, then dogs can fly.
d) If $2 + 2 = 4$, then $1 + 2 = 3$.

Answer.

a) True; $1 + 1 = 3 \equiv \perp$, unicorns exist $\equiv \perp$, $\perp \to \perp = \top$
b) True; $1 + 1 = 3 \equiv \perp$, dogs can fly $\equiv \perp$, $\perp \to \perp = \top$
c) False; $1 + 1 = 2 \equiv \top$, dogs can fly $\equiv \perp$, $\top \to \perp = \perp$
d) True; $2 + 2 = 4 \equiv \top$, $1 + 2 = 3 \equiv \top$, $\top \to \top = \top$

Problem 6. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a) Experience with C++ or Java is required.
b) Lunch includes soup or salad.
c) To enter the country you need a passport or a voter registration card.
d) Publish or perish.

Answer.

a) Inclusive or; we can have both knowledge with C++ and Java.
b) Exclusive or; the restaurant only let you choose one for lunch.
c) Inclusive or; you can present both to enter the country.
d) Exclusive or; you can't do both at the same time.

Problem 7. State the converse, contrapositive, and inverse of each of these conditional statements.

a) If it snows tonight, then I will stay at home.
b) I go to the beach whenever it is a sunny summer day.
c) When I stay up late, it is necessary that I sleep until noon.

Answer.

a) **Converse**: If I stay at home, it will snow tonight.
Contrapositive: If I don't stay at home, it won't snow tonight.
Inverse: If it doesn't snow tonight, then I won't stay at home.
b) **Converse**: Whenever I go to the beach, it is a sunny summer day.
Contrapositive: If I don't go to the beach, then it is not a sunny summer day.
Inverse: If it is not a sunny summer day, then I don't go to the beach.
c) **Converse**: Whenever I sleep until noon, it is necessary that I stay up late.
Contrapositive: Whenever I don't sleep until noon, it isn't necessary that I stay up late.
Inverse: When I don't stay up late, it isn't necessary that I sleep until noon.

Problem 8. How many rows appear in a truth table for each of these compound propositions?

a) $(q \to \neg p) \lor (\neg p \to \neg q)$
b) $(p \land \neg t) \land (p \lor \neg s)$
c) $(p \to r) \lor (\neg s \to \neg t) \lor (\neg u \to v)$
d) $(p \land r \land s) \lor (q \land t) \lor (r \land \neg t)$

Answer.

a) 4 rows for 2 propositions q, p (2 bits, 2^2)
b) 8 rows for 3 propositions p, t, s (3 bits, 2^3)
c) 64 rows for 6 propositions p, r, s, t, u, v (6 bits, 2^6)
d) 32 rows for 5 propositions p, q, r, s, t (5 bits, 2^5)

Problem 9. Construct a truth table for $((p \to q) \to r) \to s$.

Answer.

p, q, r, s	$p \to q$	$(p \to q) \to r$	$((p \to q) \to r) \to s$
$\bot\bot\bot\bot$	$\top$	$\bot$	$\top$
$\bot\bot\bot\top$	$\top$	$\bot$	$\top$
$\bot\bot\top\bot$	$\top$	$\top$	$\bot$
$\bot\bot\top\top$	$\top$	$\top$	$\top$
$\bot\top\bot\bot$	$\top$	$\bot$	$\top$
$\bot\top\bot\top$	$\top$	$\bot$	$\top$
$\bot\top\top\bot$	$\top$	$\top$	$\bot$
$\bot\top\top\top$	$\top$	$\top$	$\top$
$\top\bot\bot\bot$	$\bot$	$\top$	$\bot$
$\top\bot\bot\top$	$\bot$	$\top$	$\top$
$\top\bot\top\bot$	$\bot$	$\top$	$\bot$
$\top\bot\top\top$	$\bot$	$\top$	$\top$
$\top\top\bot\bot$	$\top$	$\bot$	$\top$
$\top\top\bot\top$	$\top$	$\bot$	$\top$
$\top\top\top\bot$	$\top$	$\top$	$\bot$
$\top\top\top\top$	$\top$	$\top$	$\top$

Challenge 1.

Fuzzy logic is used in artificial intelligence. In fuzzy logic, a proposition has a truth value that is a number between 0 and 1, inclusive. A proposition with a truth value of 0 is false and one with a truth value of 1 is true. Truth values that are between 0 and 1 indicate varying degrees of truth. For instance, the truth value 0.8 can be assigned to the statement "Fred is happy," because Fred is happy most of the time, and the truth value 0.4 can be assigned to the statement "John is happy," because John is happy slightly less than half the time.

Question. The truth value of the conjunction of two propositions in fuzzy logic is the minimum of the truth values of the two propositions. What are the truth values of the statements "Fred and John are happy" and "Neither Fred nor John is happy"?

Answer.

Let:

- $p \equiv$ Fred is happy $= 0.8$
- $q \equiv$ John is happy $= 0.4$

Since the truth value would be $\min(p, q)$ for the conjunction of propositions p and q, to get the truth value of $p \land q$, we just do $\min(0.8, 0.4)$, which is 0.4.

For the second part, we want to get the truth value of $\neg p \land \neg q$. Since fuzzy logic is from 0 to 1 inclusive, we can just subtract the truth value from 1 to get the inverse for the $\neg$ operation.

$$\begin{aligned}\neg p &\equiv 1 - 0.8 = 0.2\\ \neg q &\equiv 1 - 0.4 = 0.6\\ \neg p \land \neg q &\equiv \min(\neg p, \neg q) = \min(0.2, 0.6) = 0.2\end{aligned}$$

$$\boxed{\begin{aligned} p \land q &= 0.4\\ \neg p \land \neg q &= 0.2\end{aligned}}$$

Challenge 2. Is the assertion “This statement is false” a proposition?

Answer. The assertion “This statement is false” is not a proposition, since a proposition requires it to be either true or false (or in fuzzy logic, a number from 0 to 1 inclusive).

If the statement is false, then what it states must be false, leading to a contradiction; and if the statement is true, then the statement will also contradict itself since it states that itself is false.